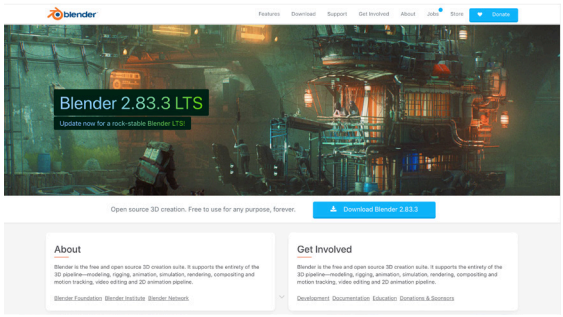
Lab: Blender Basics

Blender is a powerful tool for creating, animating and exporting 3D models and animation. The program can be a bit confusing, so this lab will take us through the basics of the program.

Blender

If you have not already you'll need to download and install the program Blender. Blender can be downloaded for free from their website blender.org. Blender works for all major desktop operating systems.

http://www.blender.org



This is the offical website for the program and a good place to get more information about the program and professionals and artists using the program.

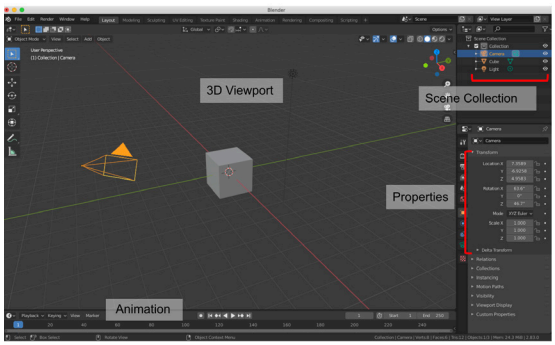
Please note that Blender is meant to have a three button mouse (a left button, a right button and a scroll wheel). If you do not have access to a three button mouse please contact me.

Begin

After the program is installed begin by loading the program. If this is the first time loading the program you'll get a little '*Welcome*' screen. Begin by opening *Blender-Basics.blend*

Workspace

The first thing you'll notice is that Blender has A LOT of stuff. We are not going to cover everything about Blender in this class, just items related to 3D modeling. Your program will most likely look something like mine on the right.



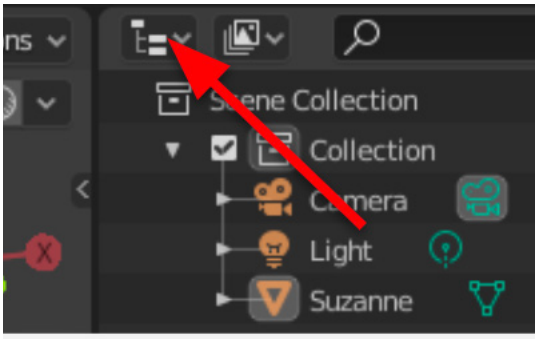
The large 3D plane area is called the *3D Viewport*. This shows us our 3D world. Think of it like the canvas in Adobe Photoshop.

The panel on the upper right is the *Scene Collection*. The *Scene Collection* lists all of the 3D world items in our document including models, camera and lights. The *Scene Collection* allows us to group items and turn them on and off. Think of the *Scene Collection* like the Layers Panel in Adobe Photoshop.

The panel on the lower right is the *Properties* panel. It shows us information about the 3D item that is selected. Think of the Properties panel like the *Properties* panel in Adobe InDesign. It

updates depending on which item is selected.

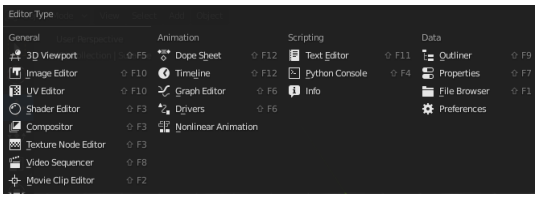
The panel on the bottom is the animation panel. This is where you would animate a 3D model. We are not going to discuss animation in this class.



Switch Panels

Everything in Blender is panel based. If you click on upper left hand corner icon of a panel it will open a dialogue box. This dialogue box will list all of the major panels including: 3D Viewport, Scene Collection, Properties, and Preferences.

Go the Scene Collection panel. Click on the icon in the top left and choose Preferences. You will see that the Scene Collection Panel has been replaced by the Preferences Panel.



If you want to make a panel take up the entire program use *Control /CTRL and spacebar*. To return back to normal use *Control/CTRL and spacebar* again.

Navigate

Just like navigating around an image in Adobe Photoshop is important, navigating around an world in Blender is important as well.

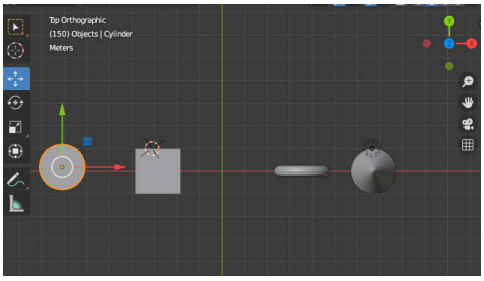
The first way to navigate is to orbit. To orbit use click and hold your middle mouse button (scrollwheel). You'll notice that you are orbitng around your default cube like a camera. Mousing your mouse clockwise or counter-clockwise will orbit clockwise or counter clockwise accordingly.

To zoom in and out of your sceen use the scroll wheel on your mouse. Scrolling up will zoom out, scrolling down will zoom in. You can also use *command / Alt + (-/+)* to zoom in an out using the keyboard. Blender also provide a zoom tool on the upper right hand side of the 3D Viewport that would allow you to zoom in and out as well. Simply click on the icon to zoom in or out.

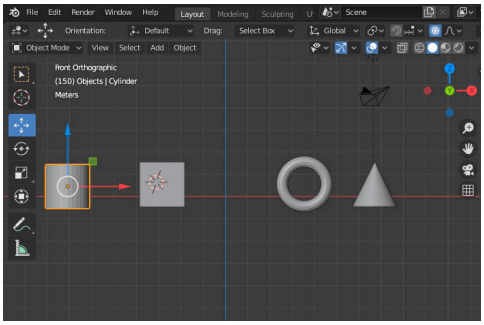
Finally we have the pan option. Panning moves the camera without orbiting or zooming. To pan use hold down the middle mouse button and the shift key on your keyboard. You can also use the hand tool on the upper right toolbar to pan around your scene as well.

Orthographic Views

Orthographic views are ways of viewing an items from a flush 90˚ perspective. Typically orthagraphic views are from common sides such as front, back, top, bottom, left and right. Orthographic views can be useful when modeling complicated objects.



To switch to an orthagraphic view we are going to use the gizmo that is in the upper right hand corner of the 3D Viewport. The gizmo can be used to orbit, but it also can be used to quickly switch to orthagraphic views. You'll notice that gizmo allows you to select axises like the X,Y or Z axis. Click on the blue circle that has the letter 'Z' on it. You'll see that the camera jumps above the cube. Suddenly our 3D world turns in a flat 2D world. At the upper left hand corner it will say Top Orthographic, meaning we are viewing our 3D object from the direct top. Click on the blue circle with the Z again and it will switch to the Bottom Orthographic view.



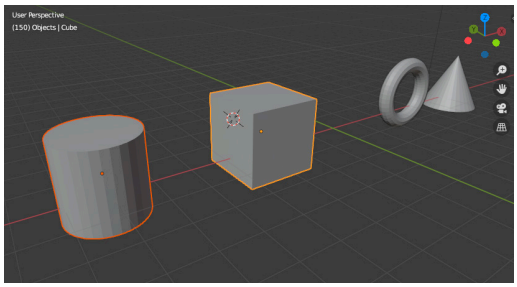
You can click on the X circle to see the right and left orthographic views and Y circle to see the front and back. Try switching around orthographic views using the Blender *gizmo*.

Toolbar

Okay, now that we can navigate let's start to manipulate our default cube. On the left side of the 3D Viewport are some basic tools to help use that. The toolbar can be opened and closed by using the *'T'* key on the keyboard.

Select

Just like we needed to select pixels in Adobe Photoshop to manipulate them we need to select an object in Blender to manipulate them. To select make sure the select option is highlighted in the toolbar. You can select an object by clicking on it. You'll know it is highlighted when a yellow halo goes around the object.

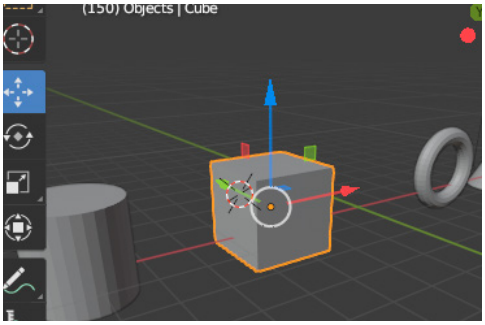


If you want to highlight multiple objects hold down the shift key on your keyboard while highlighing.

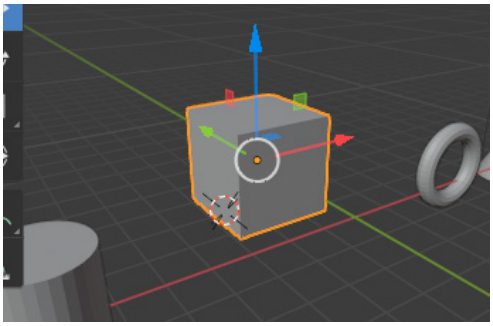
To select every object use the 'A' key on your keyboard. To deselect everything use option / ALT + A.

Move

Let begin by moving objects in our scene. Highlight the cone model in the scene. Then choose the move tool in the toolbar. You'll notice colored arrows coming from the cube. You can select an arrow to drag the cube along axises. Red is for the x-axis, green is for the y-axis and blue is for the z-axis.

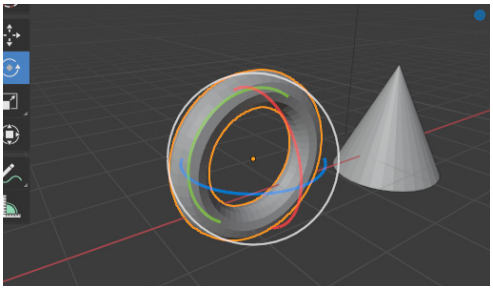


Select the blue arrow or z-axis and drag the cube up. You'll notice it move. Now select the x-axis and do the same you'll notice it move as right/left or front/back.

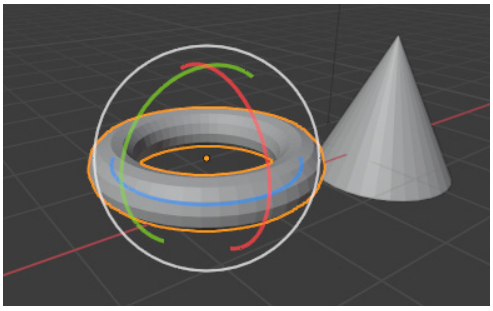


Rotate

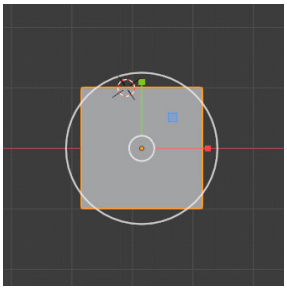
Now that we can move an object, let's rotate an object. Highlight the ring and choose the rotate option from toolbar. You'll see that the ring has red, green and blue rings around the model. These rings allow you to rotate the model along the x, y or z axis. Red for the x-axis, green for the y-axis and blue for the z-axis.



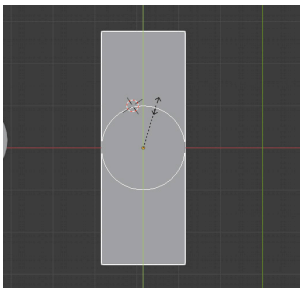
Use the rotate tool to rotate the ring roughly 90˚ on the x-axis so the ring is facing the ground plane.



Scale

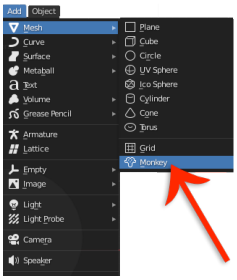


Now that we can move and rotate an object, let's learn how to scale a model. Use the gizmo to go into top orthographic mode. Choose the scale option from the toolbar. You'll notice that the arrows are replaced with small cubes.

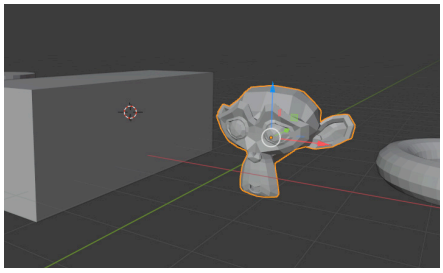


Once again the cubes colors correspond with the axises. Let's scale our cube along the y-axis.

Add an Object

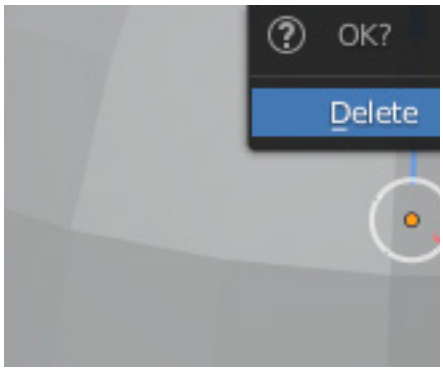


Blender allows you to quickly add basic 2D and 3D models to our world. To add a model go to the '*Add*' option in the 3D Viewport and choose *Mesh*. Then choose from the list of basic 3D mesh objects. For this lab let's choose monkey! The monkey is Susanne, who is Blender's official mascot.



Use the move, rotate and scale to place Susanne in your 3D world.

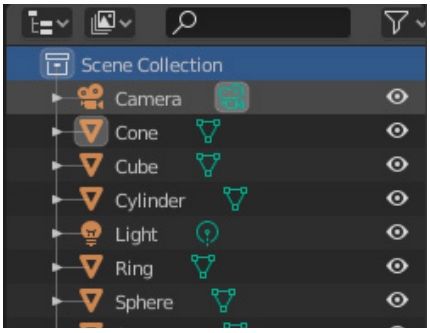
Delete an Object



Just as we may want to add an object, we may also want to delete an object as well. To delete an object from the 3D world highlight the model and press 'X' and choose 'Delete'. You can also right-click on the object and choose 'Delete' or go to Object > Delete in the menu.

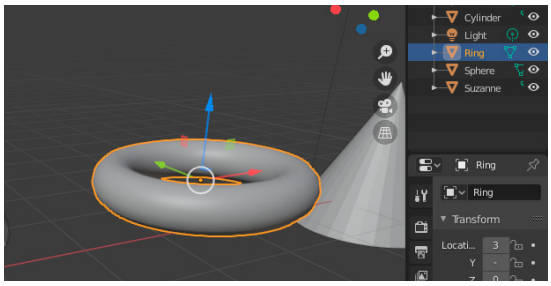
Select and delete the sphere in the scene.

Scene Collection

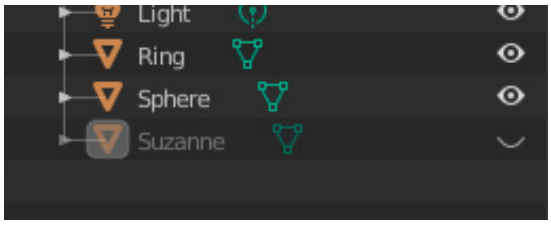


When you have a complicated 3D world there are many model and aspects to keep together. These include different 3D models, lights, cameras and other aspects of the 3D world. To assist we have a panel called the Scene Collection panel.

If you have not already done so change the *Preference* panel back into the Scene Collection panel. Take a look at the scene and you see different cames and icons. The orange triangle means the item is a 3D model. The light bulb icon means the item is a light within the 3D model and the film camera icon means that the item is scene camera.

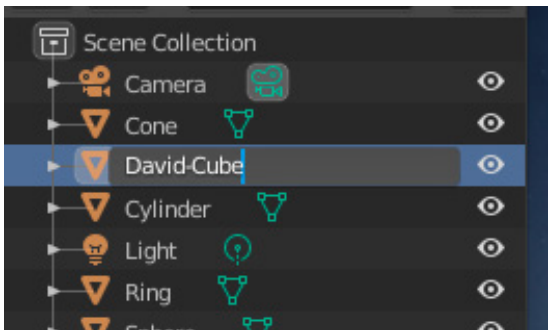


We can use the scene collection to assist with selecting items in our 3D world. Highlight the item marked '*Ring*'. Then look in your scene and you should see the 3D ring model highlighted with a yellow halo.

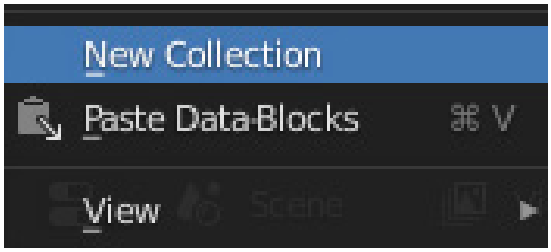


Similar to layers in Adobe Photoshop we can turn 3D objects visibility on and off using the visibility eye icon. Use visiblility icon to turn off the Suzanne object.

Finally we can use the Scene Collection menu to help us organize items. One way to do this is to rename objects. When you add an object will get a default name like *'Cube'* or *'Sphere'*, However it might be easier to name objects for ourselves. To do this we'll use the Scene Collection.

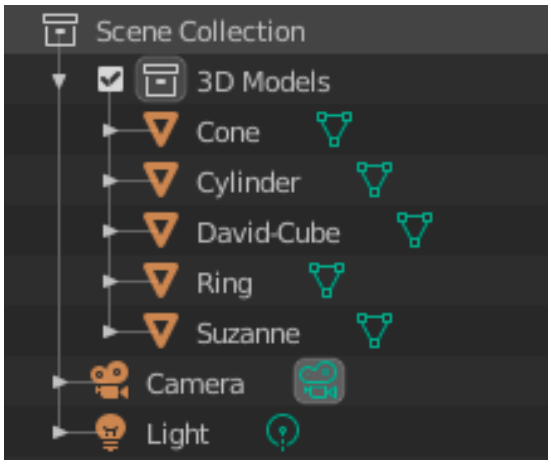


Double click on the '*Cube*' layer and rename it to '*Yourname-Cube*'. Hit Return or Enter to save the name change. You'll see that the cube name has been changed in the *Scene Collection*. If you select the cube in the scene you'll see the new name for it appear in the information in upper left hand side of the 3D Viewport.

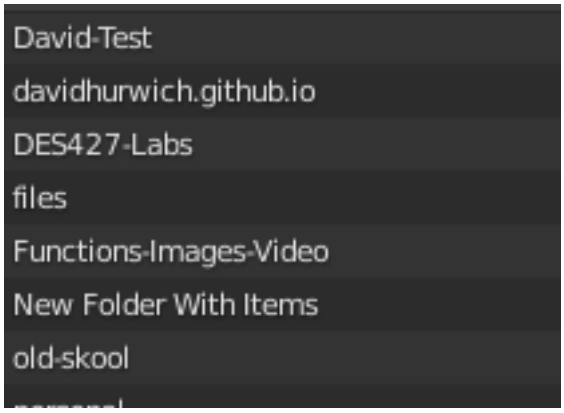


When you have a complicated file you may have many objects that you want to group together. In Blender this is called a collection. To create a new collection right-click anywhere inside the Scene Collection menu to create a dialogue box. Choose the option that says '*New Collection*'.

To rename this new collection double click on it and call it '*3D Models'*. Hit *Return / Enter* to save the renaming. Now that we've created a collection let's add objects to it. Drag the *Cone*, *Ring*, *Cylinder*, *Yourname-Cube* and *Suzanne* into the '*3D Models*' collection. You will be able to tell that the 3D models were added because they become indented.



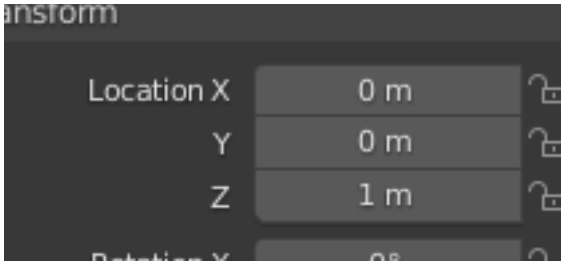
Create / Save a New Blend File.



Many times you'll need to be to create a new Blender file in order to create and modify a new scene. To create a new Blender file go to *File > New > Genera*l. You should a new 3D world appear that only have one cube.

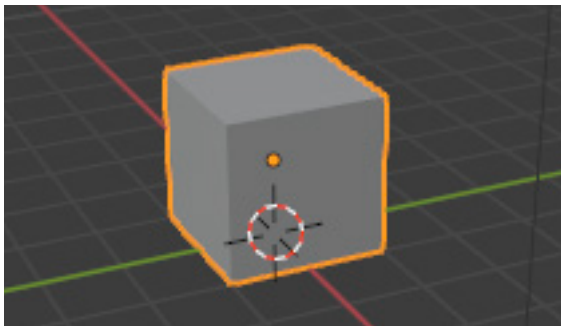
Save your new document by going to *File > Save*. The native format of Blender is the *.blend* file. Use the file manager to save your file, I recommend the Desktop. Label your Blend file *lastname-first-file.blend*.

Properties Panel



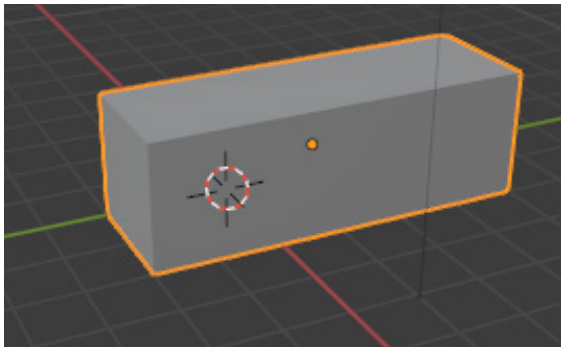
Sometimes we want precise control over an object. For that we can use the Properties panel. Make sure your *Properties Pane*l is available and that you have it set to '*Object Properties*'. Object properties is in the center of the menu that has a square icon.

With the object properties panel highlighted you should see an option marked transform. Transform allows you to make specific changes to your object.



Highlight the cube and you should see that the cube location is set to (0,0,0) or the center of the world. We can use this properties panel to set the exact location we want. With the cube highlighted set the Z value of the cube to 1. You should see you cube move to the 1 on the front orthographic. Awesome!

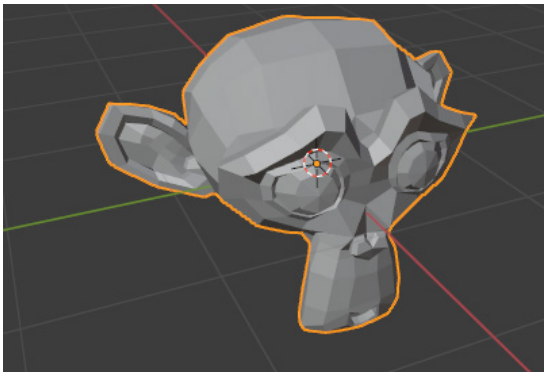
Now lets set the X and Y to 1 as well. You'll see that the cube has been moved from the center of the world to offcenter.



Now that we've moved our model let's scale our object as well. By default our cube is 1 x 1 x 1 Blender units. These are not inches or pixels by a relative measurement. Let's make the depth of the cube longer than it's width and height. Set the Scale Y to 3 and you should see the depth of your cube expand.

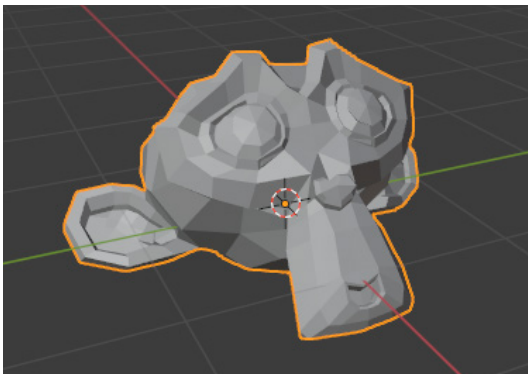
Add Suzanne

For the last part let's rotate an object. Rotation is easier to see when the model is a bit more complex so let's add the monkey or Suzanne. Go to *Add > Mesh > Monkey* and add the monkey head.



Select the monkey and look at it's rotation in the *Transform* properies.Use *Rotate Z* and set the value to 90˚. This will make the make rotate 90˚ to the right. Typically *Rotate Z* rotates left and right. To have the monkey rotate towards or away from the camera use the *Rotate X* value.

Highlightt the *Rotate X* and set it to -45˚ This will tilt the monkey head up towards the top of the document.



Lab Submission

Whew! That was a lot, but hopefully you have a good understanding of the basics of Blender.

Create a folder with your name on it and submit both Blender-Basics.blend and lastname-first-file.blend. Please make sure to zip the file and upload it to iLearn to the appropiate spot.